

Orbit — Full Budget & Cost Estimation Plan (INR)

K-12 multi-persona SPA · Vercel + Supabase + Gemini · India Pvt Ltd setup / ROC / GST / DPDP · Planning estimates ·
FX ₹83/USD · 2025–26 market bands

Disclaimer

These are planning estimates derived from public cloud list prices and typical India 2025–26 professional-fee bands. They are **not** vendor quotes, legal advice, tax advice, or accounting advice. FX assumption: **₹83 per USD**. Confirm rates with CA/CS and vendors (vercel.com/pricing, supabase.com/pricing, ai.google.dev) before school-district commitments.

1. Exclusions

Excluded from all mid totals: employee salaries, founder draw, product engineering contractors, office rent, co-working, utilities, and furniture.

Included: tech infrastructure (Vercel / Supabase / Gemini / SMS / observability) + India Year-0 incorporation & licences + annual statutory/compliance (CA/CS retainers, founder-only email 1–2 seats, accounting SaaS). Optional product contractors: **₹0** mid (excluded); project help **₹2L–₹15L/yr** is separate from statutory retainers. Razorpay MDR is pass-through (% of GMV), not fixed OpEx.

Headline — Managed Pro + India setup/compliance (ex-payroll)

Year-0 setup mid **₹1.8L** · annual compliance mid **₹1.35L**. Cash Year-1 mid (infra + compliance + Y0): 1k **₹3.7L** · 10k **₹4.9L** · 50k **₹11.3L** · 100k **₹18.1L**. AWS is **not required** at ≤50–100k MAU unless residency / VPC / RFP mandates it.

Key assumptions

- MAU = unique authenticated users/month (Student / Parent / Teacher / School).
- DAU/MAU ≈ 22% → 1k MAU ≈ 220 DAU; 10k ≈ 2.2k; 50k ≈ 11k; 100k ≈ 22k DAU.
- Sessions ≈ 1.4 / DAU / day; most reads/writes go to Supabase (RLS), not Vercel.
- Vercel load: /api/gemini, /api/notify, optional /api/razorpay/* — not every page view.
- AI blended mid: ~0.4 queries / DAU / day; ~900 in + ~450 out tokens (tutor/quiz text).
- Year-0 mid ~**₹1.8L**; annual compliance mid ~**₹1.35L** (ex-VAPT in mid).

2. Year-0 setup (one-time ₹)

Private Limited — most common for funded/SaaS. Low / mid / high are realistic 2025–26 India bands (state stamp duty moves the high end). Planning mid ≈ sum of item mids.

Year-0 band: Low **₹75,000** · Mid (plan) **₹1.8L** · High **₹4.2L**. Amortized view: **₹60,000/yr** over 3 years — cash Year-1 still needs the full outlay.

Item	Tier	Low	Mid	High	Notes
Incorporation (SPICe+, MoA/AoA, DIN, DSC×2, stamp, ROC, CA/CS)	Mandatory	₹15,000	₹30,000	₹55,000	Stamp duty state/capital-dependent
PAN / TAN	Mandatory	₹0	₹2,000	₹5,000	Usually bundled with SPICe+
GST registration	Near-mandatory	₹0	₹3,000	₹8,000	Portal free; CA help common
Bank current account opening	Near-mandatory	₹0	₹1,000	₹5,000	KYC time is the cost
Startup India DPIIT recognition	Recommended	₹0	₹5,000	₹15,000	Mostly free; CA packaging
MSME / Udyam registration	Optional	₹0	₹0	₹2,000	Free self-serve
Trademark — Orbit brand (1 class + attorney)	Recommended	₹10,000	₹18,000	₹35,000	Renewal ~every 10 years
Domain (.in / .com) — first year	Near-mandatory	₹1,000	₹2,500	₹5,000	Count once vs infra line
Founder tooling — email 1–2 admin seats (Y0)	Recommended	₹4,000	₹8,000	₹15,000	Not employee seats
Legal basics — Terms, Privacy, DPDP notices	Recommended	₹25,000	₹60,000	₹1.5L	Commercially expected
DPDP / children's data — edtech advisory	Recommended	₹15,000	₹40,000	₹1L	Consent, retention, processors
Payment gateway setup (Razorpay etc.)	Optional	₹0	₹0	₹5,000	MDR is % of GMV
Accounting software (Zoho Books / Clear) — Y0	Recommended	₹5,000	₹10,000	₹25,000	SaaS fee only

3. Annual compliance (₹ / year)

Recurring non-people costs after incorporation. Mid excludes full VAPT; high includes pen-test and richer advisory. **Band:** Low ₹45,000 · Mid (plan) ₹1.4L · High ₹3.5L.

Item	Tier	Low	Mid	High	Notes
ROC filings — AOC-4, MGT-7, DIR-3 KYC, AGM-related (CA/CS)	Mandatory	₹15,000	₹35,000	₹80,000	Statutory for Pvt Ltd
GST return filing — professional fees	Near-mandatory	₹8,000	₹24,000	₹60,000	Depends on turnover
Income-tax return / tax-audit professional fees	Near-mandatory	₹10,000	₹25,000	₹75,000	Audit trigger is receipts-based
Founder email 1–2 seats (ongoing)	Recommended	₹4,000	₹8,000	₹15,000	Exclude full team Workspace
Accounting software (annual)	Recommended	₹5,000	₹10,000	₹25,000	Zoho Books / Clear class
Domain renewal	Near-mandatory	₹1,000	₹2,000	₹4,000	One domain line
DPDP light annual advisory / policy refresh	Recommended	₹0	₹15,000	₹50,000	As school footprint grows
Cyber / professional indemnity insurance	Optional	₹0	₹15,000	₹50,000	Often expected by schools
Basic VAPT / pen-test (annual)	Optional	₹0	₹0	₹2.5L	Before larger enterprise deals

Tax audit under Income-tax rules depends on turnover/receipts — not automatic at incorporation. Your CA confirms whether audit applies in a given year.

4. Scale cases — Combined Year-1 (Managed Pro)

Cash Y1 mid = tech infra mid + annual compliance mid (■1.35L) + Year-0 mid (■1.8L). Operating mid = infra + compliance only (ex-Y0). P80 buffers AI/SMS + optional insurance/VAPT upside.

Scale	Tech mid	Compliance	Year-0	Operating	Cash Y1 mid	P80	Mo. op mid
Pilot (1k)	■55,000	■1.4L	■1.8L	■1.9L	■3.7L	■4.8L	■16,000
Early (10k)	■1.7L	■1.4L	■1.8L	■3.1L	■4.9L	■6.5L	■25,750
Growth (50k)	■8.1L	■1.4L	■1.8L	■9.5L	■11.3L	■16.5L	■79,000
Scale (100k)	■14.9L	■1.4L	■1.8L	■16.3L	■18.1L	■28L	■1.4L

Examples

Early 10k: Tech ■1.7L + compliance ■1.35L = operating ■3.1L; + Y0 ■1.8L → cash Y1 mid ■489,000 · P80 ■6.5L.
Growth 50k: Tech ■8.1L + compliance ■1.35L = operating ■9.5L; + Y0 ■1.8L → cash Y1 mid ■1,128,000 · P80 ■16.5L.

Tech infra only — Managed Pro (Year-1 ■)

Scale	MAU	Low Y1	Mid Y1	High Y1	USD mid (ref)
Pilot	1,000	■25,000	■55,000	■1.4L	~\$663
Early	10,000	■1L	■1.7L	■3.5L	~\$2,096
Growth	50,000	■4.6L	■8.1L	■14.9L	~\$9,795
Scale	100,000	■8.3L	■14.9L	■29.1L	~\$18,000

Usage model snapshot

Metric	1k MAU	10k MAU	50k MAU	100k MAU
DAU (22% of MAU)	220	2,200	11,000	22,000
Sessions / day	~310	~3,100	~15,400	~30,800
Supabase API / day (=25/session)	~8k	~77k	~385k	~770k
AI queries / day (mid)	~90	~880	~5,000	~10,000
AI tokens / mo (mid)	~4M	~40M	~220M	~440M
DB size end Y1 (mid)	0.5–2 GB	2–6 GB	8–25 GB	20–50 GB
File storage mid	1–5 GB	5–20 GB	30–120 GB	60–250 GB

5. Platform cases & AWS verdict

Mid-case Year-1 tech TCO by platform. India Year-0 + annual compliance stack on top of any choice (~■3.1L mid in Year-1 cash). **Managed Pro** is the recommended default.

Platform	1k mid	10k L / M / H	50k L / M / H	100k mid
A · Lean Hobby→Pro	■35,000	■60,000 / ■1.2L / ■2.4L	■2.7L / ■4.5L / ■9.1L	■8.3L
B · Managed Pro ★	■55,000	■1L / ■1.7L / ■3.5L	■4.6L / ■8.1L / ■14.9L	■14.9L
C · Managed Team	■7.2L	■7.2L / ■8.9L / ■12L	■11L / ■15.5L / ■24L	■23L
D · AWS mid	■5.0L	■5.0L / ■10.0L / ■19.9L	■11.6L / ■23.2L / ■45.6L	■41.5L
E · AWS enterprise	■14.9L	■16.6L / ■29.9L / ■49.8L	■33.2L / ■58.1L / ■99.6L	■91.3L
F · GCP / Railway·Fly	■83,000	■1.5L / ■2.7L / ■5.0L	■5.0L / ■9.1L / ■16.6L	■16.6L

Platform notes: A — upgrade only when limits force it. B — Vercel Pro + Supabase Pro (recommended Y1). C — SSO, SOC2, longer SLAs. D — Amplify/CloudFront + Lambda + RDS/Aurora or ECS + Cognito. E — Multi-AZ, WAF, light multi-region DR. F — more control than BaaS; lighter ops than full AWS (■2.7L @10k · ■9.1L @50k) — usually not worth rebuilding Auth/RLS in Year 1.

AWS needed? by scale

Scale	MAU	AWS needed?	Why
Pilot	1,000	No	Hobby/Free or Pro is ample; AWS adds cost and delay

Scale	MAU	AWS needed?	Why
Early	10,000	No	Managed Pro handles SPA + Postgres + Auth comfortably
Growth	50,000	No	Still within Pro unless residency/procurement forces named cloud
Scale	100,000	Rarely	Usually still Pro; AWS only for compliance, VPC, multi-region DR, or RFP
AWS verdict			
Honest: AWS is usually not required at ≤50–100k MAU. Justify only for India residency/VPC, multi-region DR, named-cloud RFPs, or isolation beyond shared BaaS. Stay on Vercel + Supabase Pro; escalate cloud only on written compliance/procurement requirements.			

Upgrade triggers

Trigger	Action
Hobby bandwidth / function / build limits or need team previews	Vercel Hobby → Pro (~■1.7k/mo + usage)
Supabase Free pause risk, >500 MB DB, or need daily backups	Supabase Free → Pro (~■2.1k/mo)
DB CPU/RAM pressure, connection saturation, slow dashboards	Compute Micro → Small (~■1.2k) → Medium (~■5k)/mo
Districts require SOC2 / SSO / longer backup retention	Supabase Pro → Team (~■50k/mo) — governance, not capacity
AI spend > ~30% of monthly tech OpEx or latency SLOs fail	Cache prompts, cap queries/user, Flash-Lite only
School RFPs ask for pen-test / cyber insurance certificates	Add VAPT (■1–2.5L) + PI/cyber premium
India residency, VPC isolation, multi-region DR, or enterprise RFP	Evaluate AWS/GCP — not required by MAU alone at ≤100k

6. Other recurring tech costs & AI note

Category deep-dive — Managed Pro (annual ■)

10,000 MAU

Category	Low	Mid	High	Drivers
Hosting / CDN / serverless (Vercel)	■0	■35,000	■75,000	Hobby free → Pro ~\$20/seat/mo + usage
DB + Auth + Realtime + Storage (Supabase)	■25,000	■35,000	■75,000	Pro \$25/mo; compute; disk; egress
AI (Gemini Flash-Lite class)	■5,000	■20,000	■60,000	~\$0.10/\$0.40 per 1M in/out tokens
Transactional email / SMS / push	■2,000	■22,000	■66,000	MSG91 SMS dominates if OTP on
Domain + SSL (infra line)	■1,000	■1,500	■3,000	Overlap with Year-0 — count once
Observability (Sentry, logs, uptime)	■0	■8,000	■30,000	Sentry free→Team
Backups / PITR / DR	■0	■5,000	■20,000	Pro daily backups 7d included
Contingency on tech (~10–15%)	■10,000	■20,000	■45,000	Price/SMS/AI swings

50,000 MAU

Category	Low	Mid	High	Drivers
Hosting / CDN / serverless (Vercel)	■40,000	■1L	■2.3L	Fluid CPU/memory; CDN quotas
DB + Auth + Realtime + Storage (Supabase)	■50,000	■1L	■3.0L	Micro→Small/Medium; disk >8 GB
AI (Gemini Flash-Lite class)	■30,000	■91,000	■3.0L	Vision/OCR = high band
Transactional email / SMS / push	■7,000	■87,000	■3.0L	MSG91 is volatile
Domain + SSL (infra line)	■1,000	■1,500	■5,000	Count once
Observability (Sentry, logs, uptime)	■10,000	■33,000	■1L	Optional log drain
Backups / PITR / DR	■0	■17,000	■60,000	PITR / longer retention optional
Contingency on tech (~10–15%)	■40,000	■90,000	■1.8L	In platform L/M/H bands

Adjacent / GTM-light (still ex-payroll, ex-office)

Cost	Low	Mid	High	Treatment
GTM-light (content, ads, school events)	■0	■50,000	■2.5L	Optional — not in cash Y1 mid
Product contractors (optional soft)	■0	Excluded	■2–15L	Separate from CA/CS retainers
Payment MDR (Razorpay)	—	% of GMV	—	Pass-through; not fixed OpEx
Contingency on tech	~10%	~12%	~15%	Baked into platform bands

AI sensitivity band (Gemini) — Year-1 ■

Volatile OpEx line after SMS. Prefer gemini-flash-lite · cap queries per student/day · cache syllabus prompts · avoid vision/OCR as default. At 50k heavy AI alone can approach ■3.0L/yr.

Scale	Light	Moderate	Heavy	Swing L→H	Note
Pilot (1k)	■2,000	■8,000	■35,000	■33,000	~0.1 / 0.4 / 2+ AI queries per DAU/day
Early (10k)	■5,000	■20,000	■60,000	■55,000	Flash-Lite text vs vision/OCR spikes
Growth (50k)	■30,000	■91,000	■3.0L	■2.7L	AI can be 10–35% of tech OpEx at heavy
Scale (100k)	■60,000	■1.8L	■5.8L	■5.2L	Cap per-user; cache prompts

Phased Year-1 (measured ramp)

Q1 front-loads Year-0 incorporation + legal/DPDP packs with pilot infra (~1k MAU). Q2–Q4 ramp schools while annual ROC/GST/CA run-rate settles. Measured ramp ends ~■3.3L cumulative operating+compliance before a dense 50k infra path (~■8L+ tech alone).

Quarter	Focus	Quarterly mid	Cumulative
Q1	Incorporate + legal/DPDP packs + pilot infra	■45,000	■45,000
Q2	Early schools, Supabase Pro, ROC/GST cadence starts	■70,000	■1.1L

Quarter	Focus	Quarterly mid	Cumulative
Q3	Growth traffic, AI/SMS monitoring	■95,000	■2.1L
Q4	Stabilize toward 10–50k path	■1.2L	■3.3L

7. Tradeoffs summary

Cost · ops burden · time-to-ship · lock-in · compliance fit · team skills — qualitative for Indian school SaaS Year 1.

Platform	Cost	Ops	Ship speed	Lock-in	Compliance	Skills
Lean Hobby→Pro	Lowest	Lowest	Fastest	Medium (BaaS)	Weak if Free pauses	Low
Managed Pro ★	Low–mid	Low	Fast	Medium	Good (Mumbai + DPA)	Low (TS + SQL/RLS)
Managed Team	High fixed	Low	Fast	Medium	Strong (SOC2/SSO)	Low
AWS mid	High	High	Slow	High (IAM/VPC)	Strong if designed	High (cloud eng)
AWS enterprise	Highest	Highest	Slowest	Highest	Strongest	Highest
GCP / Railway-Fly	Mid	Mid	Medium	Medium–high	Varies	Mid

AI + SMS volatility

Gemini spend moves with product design, not host choice. MSG91 SMS is the other India-specific spike — DLT templates and per-school broadcast limits matter more than CDN pricing. Company-law compliance cost is mostly fixed vs MAU.

Recommended Year-1 budgets to approve

Cash Y1 mid = Managed Pro tech + annual compliance (■1.35L) + Year-0 (■1.8L). P80 buffers AI/SMS and optional VAPT/insurance. Excludes salaries and office.

Scale	Cash Y1 mid	P80 buffer	Notes
Pilot 1k	■3.7L	■4.8L	Tech ■55,000 · AWS: No
Early 10k	■4.9L	■6.5L	Tech ■1.7L · AWS: No
Growth 50k	■11.3L	■16.5L	Tech ■8.1L · AWS: No
Scale 100k	■18.1L	■28L	Tech ■14.9L · AWS: Rarely

Board / founder one-liner

Approve cash Year-1 mid ■4.9L for Early (10k) and ■11.3L for Growth (50k), with P80 buffers ■6.5L / ■16.5L. That covers Managed Pro tech, Year-0 India setup (~■1.8L), and annual ROC/GST/CA/DPDP run-rate (~■1.4L) — still excluding payroll and office. Stay on Vercel + Supabase Pro; escalate cloud only on written compliance/procurement requirements.

Source of truth: Orbit canvas budget model · FX ■83/USD · planning estimates 2025–26 · not legal/tax advice. Document generated for founder / board planning.